Bachelor Thesis

Adding temporal dependencies and user-aware context to a joint prediction of Valence and Arousal from Self-Annotated Text

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Abstract

1. Introduction

2. Background

This chapter provides the theoretical foundation for the proposed architecture. It begins with an overview of sentiment analysis and its limitations as a lexical or coarse-grained task, followed by the dimensional emotion modeling as a more psychologically grounded alternative. Subsequently, the transformer-based pre-trained language model BERT is presented, with special attention to its fine-tuning possibilities. The chapter ends with an overview of multi-output classification and regression with temporal and user-aware modeling, both of which build the foundation for the experimental design of this thesis.

2.1. Sentiment Analysis & Emotion Detection

This section explores the history of sentiment analysis (SA), its limitations as a coarse-grained representation of human emotion and the transition toward dimensional emotion modeling as a more expressive alternative.

SA, roughly defined as the computational study of opinions, attitudes and emotions expressed via written text (Liu 2020), has emerged as one of the most actively researched areas Natural Language Processing. Early approaches relied on manufactored dictionaries mapping words to sentiment scores, such as SentiWordNet or VADER. While these methods are simple to apply and computationally efficient, they are limited by their inability to capture context, negation and domain-specific language. Meta-analyses confirm that transformer-based approaches consistently outperform lexicon-based appraches across a wide range of domains (Hartmann et al. 2023).

The dominant task in SA is coarse-grained classification, assigning discrete labels such as positive, neutral and negative to a text. While useful for simple applications like reviews, this representation fails to capture the full complexity of human affect. More braodly, the transition from opinion mining to emotion detections reveals that affective states cannot be accuratly reduced to a single polarity dimenstion (Yadollahi et al. 2017).

A further limitation concerns the annotation process in existing SA datasets. Lables are typically assigned by third-party annotators rather then by the authors themself, which leads to a annotator bias and greater distance from ground truth. Furthermore, this ruslts in limited validity of the resulting models trained on such datasets (Rohde et al. 2025). This annotator-author mismatch is particularly problematic for tasks that target the subjective emotional state of individuals, as addressed in the present work.

While fine-grained classification partially addresses this limitations, binary lables are coarser then a single dimension of affect.

2.2. Dimensional Emotion Modeling

The following section introduces dimensional emotion modeling as a theoretically grounded framework that represents emotional states along two dimensions, capturing a richer and more continous picture of human affact.

Rather than assigning discrete emotion labels to a text, dimensional emotion represents the human affective states as points in a continuous multi-dimensional space. A widely used representation is Yadollahi et al. (2017), which organizes emotion along two independent axes: *valence*, describing the positivity or negativity of a emotion and *arousal*, describing the activation. This two-dimensional representation allows more fine-grained and psychologically grounded description, as any emotional state can be located in the valence-arousal space.

Posner et al. (2005) highlights the theoretical completeness of dimensional models compared to Ekman's basic emotion theory. Such approaches are limited to prototypical states, which fail to represent the full continuum of affect.

In the area of NLP, dimensional modeling has gained traction with the indtroduction of dedicated corpora, such as EmoBank Buechel and Hahn (2017), which provides 10,000 sentences annotated with valence, arousal and dominance. These resources have enabled the development of regression and classification models that predict continuous or ordinal scores, forming the methodological foundation on which the present work relies.

The increased representational complexity of dimensional emotion modeling places higher demands on the attention mechanism of the underlying text encoder.

2.3. Transformer-based Language Models

The following section introduces BERT, whose bidirectional attention mechanism provides the necessary context for every word and is particularly well-suited to meet the needs of this work. The development from recurrent neural networks to the transformer architecture published by Vaswani et al. (2017), with its self-attention mechasim, is a fundamental base for current models. Self-attention allows each token in an input sequence to influence every other token simultaneously. This enables the model to capture long-range dependencies without the vanishing gradient problems associated with prior recurrent architectures. The resulting representations are deeply contextual, i.e. the encoding of every token is a function of the entire input sequence.

Building on this foundation, Devlin et al. (2019) developed BERT (Bideirectional Encoder Reprenentations from Transformers), a pre-trained transformer-based language model trained on large text corpera. It is self-supervised and trained with two objectives. The first, Masked Language Modeling (MLM), randomly masks a subset of input tokens and trains the model to predict them from their bidirectional context. The second, Next Sentence Prediction (NSP), trains the model to determine, whether two sentences follow each other in the original text. Together, both objectives allow BERT to represent language in a context-sensitive manner without requiring labels.

A key design feature of BERT is the `CLS` token, which is prepended on every input and whose final hidden state serves as an aggregated representation of the entire input. This sentence-level embedding is used as the input for task-specifc heads during fine tuning. Fine-tuning adpats this pre-trained model to a downstream task by continuing training on a smaller but labled dataset, updating all model parameters jointly (Howard and Ruder 2018). This approach is espacially well suited for tasks with scarce labled training data, as the pretrained weights already encode linguistic context and dependencies which transfer through different tasks and domains.

For downstream tasks, a linear layer is appended to the `CLS` representation and trained together with the encoder. This layer can be fine-tuned for every specific task by adjusting, for example, learning rate, number of epochs etc., mapping the `CLS`-embedding via softmax activation to a descrete set of classes or producing a continuous scalar output via regression. Sun et al. (2019) demonstrates a significant increase of performance as a result of the fine-tuning strategy. These considerations are directly relevant for the experimental setup of this work because of the small size of the underlying dataset.

2.4. Multi-Output Classification & Regression

Furthermore, models benefit from multi-output classification and regression, which refers to the problem of predicting two or more targets simultaneously from a shared input. The straightforward approach is to train a separate model for each task individually, treating the targets as entirely unrelated. While beeing simple, this approach fails to exploit any shared structure between targets and may lead to suboptimal perfomances when the outputs are related, as shown by Borchani et al. (2015).

In the context of transformer-based NLP, Liu et al. (2019) demonstrate that a shared BERT encoder with task specific-heads can be trained jointly on NLP tasks. This approach consistently outperformed single-task fine-tuned models. This research extends to SA: Sane et al. (2019) show that even in the absance of strong label correlation, jointly training for sentiment, emotion and sarcasm yields mutual benefits across different tasks. This finding is directly consistent with the boomerang shaped, globally negligible correlation between valence and arousal observed in the present dataset for this work (see section 3.3.2).

The output heads can be configured for either classification or regression. Both formulations are evaluated in this work, as the ordinal nature of the valence and arousal scales is theoretically defensible.

2.5. Temporal & User-Aware Modeling

Standard NLP models treat each input as an individual, context-free instance. In affective computing, however, emotional states are inherently situated in time and shaped by the atuhor's induvidual differences. This section reviews recent work on incorporating temporal and user-aware modeling into NLP models.

2.5.1. Temporal Modeling

Adding temporal context has been shown to improve emotion and sentiment prediction in several recent studies. Ninalga (2023) demonstrate that prepending the year of a given timestamp to the input text increases perfomance in SA across multiple transformer-based models. The key advantage of this approach is its model-agnosticism: since the timestamp is prepended as text, it benefits from the same attention mechanisms of the encoder as the textual input, requiring no architectual modifications. Whether similar benefits extend to dimensional emotion prediction and whether finer-grained temporal signals such as time of day and collection wave provide additional signals is an open question that the present work addresses empirically.

2.5.2. User-Aware Modeling

Individual differences between authors are not captured by standard NLP models. Human emotional perception is inherently subjective: the same textual input can represent different emotional states across individuals. Ngo et al. (2022) demonstrates this empirically, showing that models, trained with awareness of the annotator identity, significantly outperform user-agnostic baselines.

3. Dataset

This chapter describes the dataset introduced in Soni et al. (2026), which served as the basis for SemEval 2026 Task 2. Following an overview of the exact task and data structure, an Explorative Data Analysis examines class distributions, label correlations, temporal patterns, user-specific information and the difference bewtween essays vs. feeling words. Each of which informs the modeling decisions handled in Chapter 4

3.1. SemEval 2026 Task2 - Task Description

The following section names the task and explains the opportunity of the Dataset connected with it.

Soni et al. (2026) formulates it as: "Given a sequence of m texts (essays or feeling words), $e_1, ..., e_m$ in chronological order, produce Valence & Arousal (V&A) scores $(v_1, a_1), ..., (v_m, a_m)$ one pair for each text [...].". The texts for this task were produced by U.S. service-industry workers that answered the question, how they were feeling.

Such self-annotated datasets are rare, as tasks of this kind are usually based on social media data e.g. tweets that are labeled by a third-party as shown by Zhang et al. (2026). These labels may provide a closer approximation to the ground truth, as it avoids the annotator bias introduced by external third-party labeling.

3.2. Data Structure

This section will name and describe all features of the dataset.

Feature	Description	type
user_id	Author Identifier	int
text_id	Text Identifier	int
text	produced text	string
timestamp	time of annotation	timestamp
collection_phase	phase of collection	int
is_words	text or feeling words	boolean
valence	Label: -2 to +2	int
arousal	Label: 0 to +2	int

3.3. Exploratory Data Analysis (EDA)

The following section will explore the data and information of the training dataset by analyizing different aspects relevant for the task.

3.3.1. Class Distribution

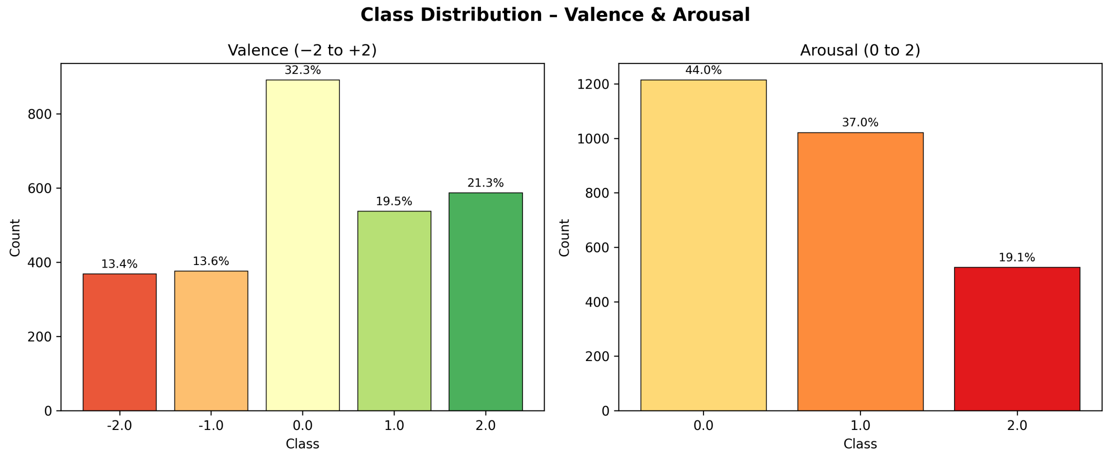

Figure 3.1.: Class distribution of valence (left) and arousal (right) labels in the SemEval 2026 Task 2 training dataset.

As shown in Figure 3.1, both target variables exhibit a notable imbalance. For valence, the neutral class (0) accounts for the largest share of scores, while the unpleasant classes (-1, -2) are underrepresented and the pleasant classes (1, 2) are approximately uniformly distributed. A similar pattern is observed for arousal, where the lowest activation (0) dominates the distribution. With increasing activation levels, the share of samples decreases monotonically, which leads to underrepresentation of the highest activation (2).

This imbalance is consistent with existing literature. Bradley and Lang (1994) report a boomerang-shaped distribution where neutral-low arousal states dominate, while neutral and high-arousal combinations are virtually absent. However the joint distribution reveals an unexpected concentration of samples in the low-arousal (0), extreme-valence (-2, 2) cells, as shown in Figure 3.2. While this partially contradicts the existing literature, this may reflect a scale-use-effect, i.e., when only three arousal levels are offered, participants may systematically assign the lowest activation level to a broader range of emotional states.

To account for this imbalance during training, class-weighted cross-entropy loss with weights inversely proportional to class frequencies, is tested.

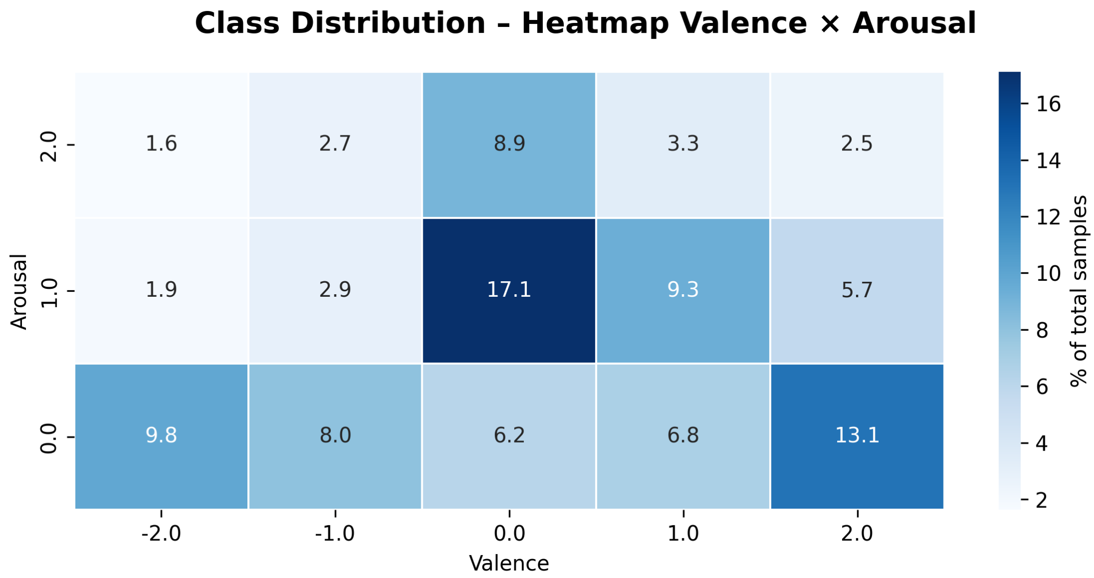

Figure 3.2.: Heatmap of valence (x axis) and arousal (y axis) pairs in the SemEval 2026 Task 2 training dataset.

3.3.2. Correlation between Valence and Arousal

While the marginal distributions characterize each target variable in isolation, Russell (1980)'s circumplex model predicts a systematic co-variation between valence and arousal. The following analysis examines whether this theoretical relationship is reflected in the present dataset.

A global Spearman rank correlation between valence and arousal yields $\rho = .017$ ($p = .386$), with Kendall's $\tau = .009$ ($p = .562$) confirming the negligible effect. At first glance, this suggests that valence and arousal are largely independent in the present dataset, which appears to contradict Russell (1980)'s circumplex model. However, this global measure obscures a meaningful non-linear structure in the data.

When the valence range is split at the neutral class (0), strong and highly significant correlations emerge in both subranges. Within the negative valence range (-2 to 0, $n = 1{,}638$), a moderate positive correlation is found ($\rho = .413$, $p < .001$; $\tau = .370$, $p < .001$), indicating that more negative valence is associated with higher arousal — consistent with the distinction between high-arousal states such as anger and low-arousal states such as sadness. Within the positive valence range (0 to $+2$, $n = 2{,}018$), the correlation reverses in direction ($\rho = -.341$, $p < .001$; $\tau = -.308$, $p < .001$), reflecting that more positive valence is associated with higher arousal, consistent with the contrast between contentment and excitement.

This V-shaped pattern is confirmed directly by a quadratic test, correlating Valence^2 with arousal, which yields $r = -.320$ ($p < .001$). The negative sign indicates that samples near the neutral class (Valence $= 0$) exhibit the lowest arousal, while extreme valence values — in either direction — are associated with higher activation levels. This result is in line with the structural prediction of Russell (1980) & the conclusion of Bradley and Lang (1994) and confirms that the global correlation of $\rho = .017$ is not evidence of independence, but rather the result of two opposing monotonic trends cancelling each other out.

Taken together, these findings provide strong empirical justification for a joint prediction architecture. The non-linear dependency between valence and arousal constitutes shared structure that a dual-head model with a common BERT encoder can implicitly exploit, even though it is invisible to models that treat the two targets as entirely independent tasks.

3.3.3. Temporal Analysis

The following subsection will analyze the long-term and short-term temporal dimension. The long-term Analyses focuses on the collection wave and publishing month, while the short-term Analysis focuses on the hour of the publication.

Long-Term Analysis

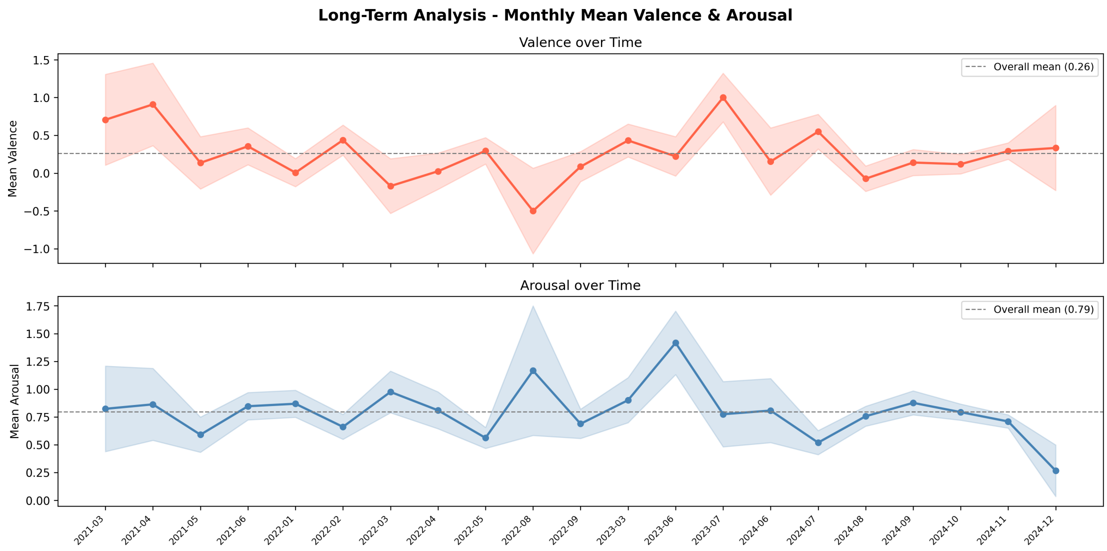

Figure 3.3.: Monthly mean valence (top) and arousal (bottom) over the full data collection period of approximately two years. Shaded areas indicate 95% confidence intervals. The dashed line represents the overall mean across all months.

The dataset spans seven collection waves over a period of approximately two years. A Kruskal-Wallis test reveals significant differences in label distributions across waves for both valence ($H = 16.255$, $p = .012$) and arousal ($H = 20.317$, $p = .002$), indicating that the emotional states reported by participants shift systematically over the course

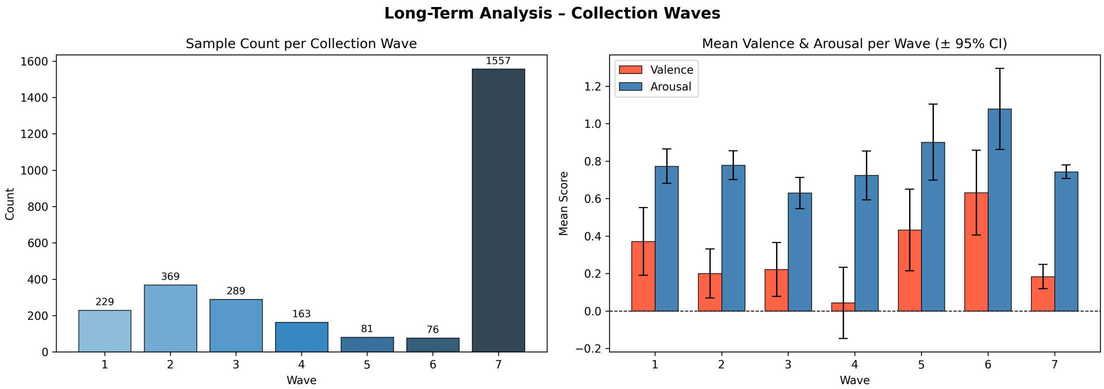

Figure 3.4.: Sample count per collection wave (left). Mean valence and arousal per collection wave with 95% confidence intervals (right).

of the study. This finding suggests that `collection_phase` carries predictive signal beyond the raw text and motivates its inclusion as an additional input feature. As shown in Figures 3.4 and 3.3, the monthly trend further reveals gradual fluctuations in mean valence and arousal over time, consistent with the presence of longitudinal affective dynamics in the data.

Short-Term Analysis

Beyond long-term trends, the *timestamp* provides information about the time of day at which a diary entry was recorded. As shown in Figure 3.5, both valence and arousal exhibit systematic variation across hours of the day. Valence peaks at 14:00 ($\bar{x} = 0.615$) and reaches its lowest mean at 10:00 ($\bar{x} = 0.017$), while arousal peaks at 18:00 ($\bar{x} = 1.000$) and is lowest at 11:00 ($\bar{x} = 0.422$). These patterns are consistent with findings from chronopsychology, where afternoon hours are associated with elevated mood and early evening with heightened activation (Stone et al. 2006).

The distribution of samples across time-of-day bins is heavily skewed toward the afternoon (12–18h, $n = 1{,}640$), followed by late night (22–24h, $n = 510$) and evening (18–22h, $n = 331$). No entries were recorded between midnight and 6:00, reflecting the naturalistic diary context of the dataset. A Kruskal-Wallis test confirms significant differences across time bins for both valence ($H = 9.672$, $p = .022$) and arousal ($H = 12.259$, $p = .007$), providing empirical support for the inclusion of timestamp-derived features in the model.

Temporal Analysis Conclusion

Taken together, both analyses provide empirical justification for incorporating temporal features into the prediction architecture. Whether encoded as a single timestamp or decomposed into separate features, such as the hour of day and the collection wave, the model may benefit from the systematic affective variation captured by these signals.

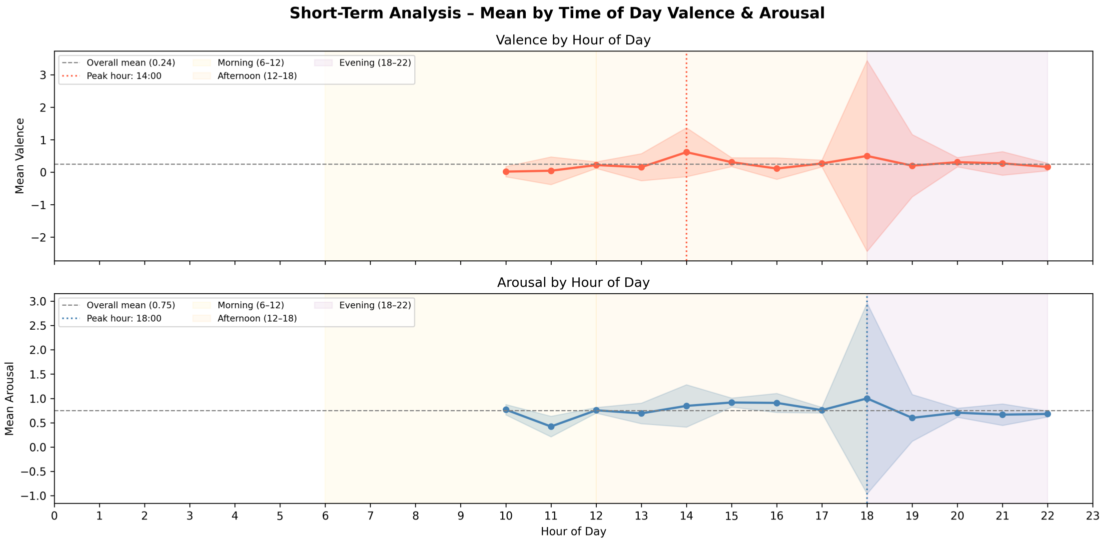

Figure 3.5.: Mean valence (top) and arousal (bottom) by hour of day. Shaded areas indicate 95% confidence intervals. Coloured regions denote morning (6–12h), afternoon (12–18h), and evening (18–22h) periods. Dotted vertical lines mark the peak hour for each dimension.

3.3.4. User-specific Analysis

The dataset contains 2,764 samples from 137 unique users, with a highly skewed distribution across users (mean = 20.2, median = 14.0, min = 2, max = 206, $SD = 28.0$). As illustrated in Figure 3.6, a small number of users account for a disproportionately large share of the data, which has direct implications for the train/test split strategy.

At the user level, mean valence ranges from -1.76 to 2.00 ($SD = 0.78$), while mean arousal ranges from 0.00 to 1.56 ($SD = 0.32$), as shown in Figure 3.7. The larger spread in per-user valence means suggests that users differ more strongly in their habitual valence reporting than in their arousal reporting. A direct variance decomposition confirms that between-user differences account for 32.6% of total valence variance and 16.3% of arousal variance, with within-user variance dominating in both cases (67.4% and 83.7% respectively). This is further supported by a Kruskal-Wallis test, yielding highly significant differences across users for both valence ($H = 823.833$, $p < .001$) and arousal ($H = 641.055$, $p < .001$).

Taken together, these findings indicate that user identity carries meaningful predictive signal for both target dimensions, motivating the inclusion of user identifiers as additional input features. At the same time, the dominance of within-user variance suggests that the model must primarily learn from the textual content of individual entries.

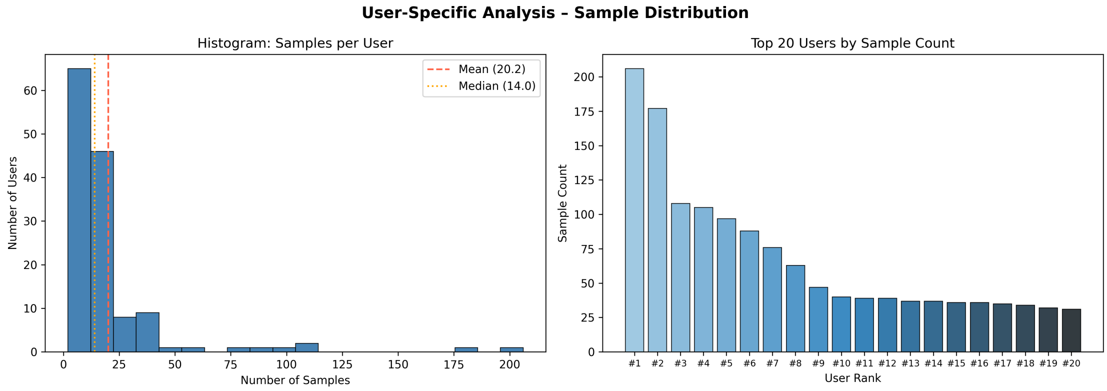

Figure 3.6.: Distribution of sample counts across users. Histogram of the number of entries per user, with mean (= 20.2) and median (= 14.0) indicated by dashed and dotted lines respectively (left). Top 20 users ranked by sample count (right).

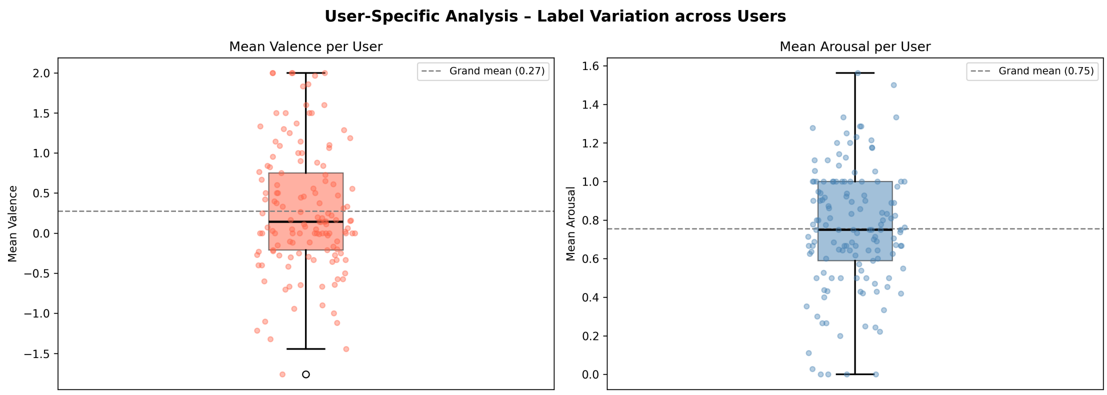

Figure 3.7.: Distribution of per-user mean valence (left) and mean arousal (right) across all 137 users, with the grand mean indicated by a dashed line (valence: 0.28; arousal: 0.76).

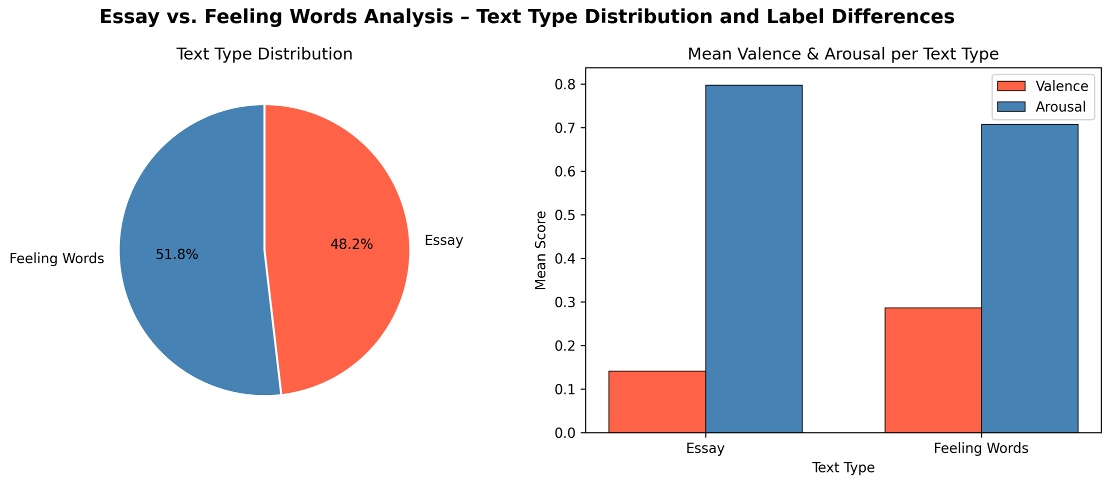

Figure 3.8.: Distribution of text types (left)(essay vs. feeling words). Mean valence and arousal per text type with 95% confidence intervals (right).

3.3.5. Essay vs. Feeling Words Analysis

The dataset is approximately evenly split between two text types: feeling words ($n =$ 1,433, 51.8%) and ecological essays($n =$ 1,331, 48.2%). As shown in Figure 3.8, the two types differ significantly in their affective content. Essays yield a lower mean valence ($\bar{x} = 0.141$) compared to feeling word entries ($\bar{x} = 0.287$), with a significant difference confirmed by a Mann-Whitney U test ($U = 891{,}053.5$, $p = .002$). For arousal, the pattern reverses: essays exhibit a higher mean ($\bar{x} = 0.798$) than feeling words ($\bar{x} = 0.708$), again significantly so ($U = 1{,}011{,}233.0$, $p = .003$).

These findings suggest that the difference between feeling words and ecological essays carries predictive signal for both target dimensions, motivating the inclusion of different treatment in the architecture, which will be further explained in the **??** chapter.

EDA Conclusion Having examined class distributions, label correlations, temporal patterns, and user-specific effects, the following paragraph summarizes the key findings and their implications for model design. The EDA reveals five key properties of the given training dataset that can be exploited for the model performance.

First, both target variables exhibit a notable class imbalance, motivating the use of class-weighted cross-entropy loss.

Second, while the global correlation between valence and arousal is negligible ($\rho =$ .017, $p = .386$), a significant Boomerang-shaped structure is present ($r = -.320$, $p <$ .001), consistent with Bradley and Lang (1994) providing empirical motivation for a joint prediction architecture.

Third, both *collection_phase* and time of day show significant variation in label distributions ($p < .05$), indicating that temporal features carry predictive signal beyond the raw text.

Fourth, a non-negligible portion of label variance is structured at the user level ($\mathrm{ICC}(1)_{\mathrm{valence}} = .264$; $\mathrm{ICC}(1)_{\mathrm{arousal}} = .198$, $p < .001$), motivating the inclusion of user identifiers as additional input features. These findings collectively justify the modeling approach pursued in Chapter 4.

Fifth, feeling words and ecological essays differ significantly in both valence ($\bar{x}$words = 0.287, $\bar{x}$essays = 0.141, $p = .002$) and arousal ($\bar{x}$words = 0.708, $\bar{x}$essays = 0.798, $p = .003$), indicating that text type itself provides predictive signal for both target dimensions.

4. Methodology

This chapter describes the complete experimental setup of this work, from data preprocessing to model evaluation. The methodology is structured as follows:

4.1. Baseline

4.1.1. Overview

The baseline model establishes the foundational architecture upon which all extended models are built. It follows the standard fine-tuning approach for transformer-based multi-output learning, without any metadata integration or architectural extensions beyond the dual-output heads. The model takes the raw input text as its sole input and simultaneously predicts valence and arousal through two independent task-specific heads, sharing a common encoder.

Two output formulations are evaluated within the baseline: a classification formulation, in which each head maps the shared text representation to a discrete set of ordered classes via cross-entropy loss, and a regression formulation, in which each head produces a continuous scalar output trained with mean squared error loss. As pooling strategy and other hyperparameters were optimized separately for each formulation, their respective configurations are reported individually in Appendix A.1. The regression formulation is additionally motivated by the class imbalance observed in Section 3.3, which it circumvents by treating the target variables as continuous ordinal values.

The baseline serves two purposes: first, it provides a performance reference for all subsequent experiments, second, it isolates the signal of the raw text, against the additional predictive value of metadata features, which can be assessed in Section **??**.

4.1.2. Preprocessing

The following section describes the preprocessing from tokenization and truncation to label encoding and the train & validation split.

Tokenization and Truncation

All input texts are tokenized using the `microsoft/deberta-base-mnli` tokenizer. Each text is truncated to a maximum sequence of 128 tokens and padded to this length using the model's padding token. The choice of 128 is motivated by a token-length analysis: the texts have a median length of 16 tokens and the 95th percentile length is 86 tokens or fewer, such that 128 tokens covers the vast majority of entries without unnecessary padding overhead. The tokenizer outputs `input_ids` and `attention_mask`, both converted to `torch.long` tensors.

Label Encoding

Valence labels are shifted by $+2$ to yield zero-indexed class indices $\{0, 1, 2, 3, 4\}$, as PyTorch's cross-entropy loss requires non-negative integer targets. Arousal labels are kept as-is, yielding class indices $\{0, 1, 2\}$. For the regression formulation, both label sets are cast to `torch.float` without further transformation, preserving the ordinal scale of the original annotations.

Train/Validation Split

The dataset is split into a training set (80%) and a validation set (20%) using a fixed random seed to ensure reproducibility. The split is applied at the sample level without user-aware partitioning, as the baseline model does not incorporate user identity as a feature.

4.1.3. Model Architecture

The following section explains the detailed encoder and pooling strategy. Furthermore, it describes the layers of the task-specific heads.

Encoder and Pooling

The encoder is initialized from `microsoft/deberta-base-mnli`, loaded via the HuggingFace `AutoModel` interface. It processes the `input_ids` and `attention_mask` and outputs the full sequence of hidden states (`last_hidden_state`) of dimension $d = 768$.

The pooling strategy is selected based on ablation results and differs between the two output formulations. For the classification formulation, the `[CLS]` token representation is used as the sentence-level representation $p \in \mathbb{R}^{768}$. For the regression formulation, mean pooling is applied over the token-level hidden states instead, weighted by the attention mask to exclude padding tokens. Formally, for a sequence of hidden states $H \in \mathbb{R}^{n \times d}$ and attention mask $m \in \{0, 1\}^n$, the pooled sentence representation is:

$$p = \frac{\sum_{i=1}^{n} m_i h_i}{\sum_{i=1}^{n} m_i} \tag{4.1}$$

where n is the sequence length and $d = 768$ is the hidden dimension. Both pooling strategies produce a sentence representation of fixed size $p \in \mathbb{R}^{768}$ regardless of input length.

A dropout layer with rate $p = 0.1$ is applied to the pooled representation for regularization during training.

Task-specific Heads

Two independent task-specific heads project the pooled representation p to their respective output spaces. The architecture of these heads differs between the two output formulations, based on ablation results.

For the classification formulation, each head consists of a linear layer with hidden dimension 256, followed by a ReLU activation and a final linear projection to the output dimension:

- **Valence Head:** Linear$(768 \rightarrow 256) \rightarrow$ ReLU $\rightarrow$ Linear$(256 \rightarrow 5)$
- **Arousal Head:** Linear$(768 \rightarrow 256) \rightarrow$ ReLU $\rightarrow$ Linear$(256 \rightarrow 3)$

Each head outputs raw logits which are passed directly to the cross-entropy loss without softmax activation, as PyTorch's `CrossEntropyLoss` applies log-softmax internally.

For the regression formulation, ablation results favored a direct linear projection without an intermediate hidden layer. Each head therefore consists of a single linear layer mapping the pooled representation directly to a scalar output:

- **Valence Head:** Linear$(768 \rightarrow 1)$
- **Arousal Head:** Linear$(768 \rightarrow 1)$

4.1.4. Training Setup

Loss Function

Both output heads are trained jointly using a combined loss. In the classification formulation, each head applies `CrossEntropyLoss`, which incorporates log-softmax internally and expects raw logits as input. In the regression formulation, each head applies `MSELoss` over the continuous scalar outputs. In both cases, the total loss is computed as the unweighted sum of the two head losses:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{valence}} + \mathcal{L}_{\text{arousal}} \tag{4.2}$$

Hyperparameters

All hyperparameters were selected through systematic ablation studies over three random seeds (42, 123, 456), following Dodge et al. (2020) for robust fine-tuning on small datasets. The final configuration is summarised in Table 4.1, with detailed ablation results reported in Appendix A.1.

Evaluation Metric

The official evaluation metric of Soni et al. (2026) is used throughout this work. For each target dimension (valence and arousal), performance is measured along two axes: between-user correlation and within-user correlation.

Hyperparameter	Classification	Regression
Encoder	`microsoft/deberta-base-mnli`	
Pooling	`[CLS]`	Mean
Head Hidden Dimension	256	None
Learning Rate	2×10^{-5}	2×10^{-5}
Batch Size	16	16
Dropout	0.1	0.1
Seeds	42, 123, 456	

Table 4.1.: Final hyperparameter configuration for the classification and regression baseline, selected via separate ablation studies.

A direct comparison between the two formulations of the baseline is presented in Chapter **??**. The regression formulation achieves a higher composite correlation overall ($r_{\text{composite}} = 0.589 \pm 0.004$ vs. 0.585 ± 0.008) and outperforms classification on valence ($r_{\text{valence}} = 0.657 \pm 0.003$ vs. 0.649 ± 0.017), while both formulations achieve virtually identical performance on arousal ($r_{\text{arousal}} = 0.521$ for both).

Between-User Correlation For each user u, the mean predicted score and mean gold score are computed across all texts associated with that user. Pearson's r is then calculated across users using these per-user means:

$$r_{\text{between}} = r\left(\left\{\underset{t\in u}{\text{mean}}\,\hat{y}_{u,t}\right\}_{u=1}^{N}, \left\{\underset{t\in u}{\text{mean}}\,y_{u,t}\right\}_{u=1}^{N}\right) \tag{4.3}$$

Within-User Correlation For each user u Pearson's r is computed between the predicted and gold scores across that user's texts. The per-user correlations are then averaged across all users:

$$r_{\text{within}} = \underset{\forall u}{\text{mean}}\left(r_u\left(\{\hat{y}_{u,t}\}_{t\in u}, \{y_{u,t}\}_{t\in u}\right)\right) \tag{4.4}$$

Composite Correlation The two correlation measures are combined into a single composite score using Fisher's z-transformation, which stabilises the variance of Pearson's r prior to averaging:

$$r_{\text{composite}} = \tanh\left(\frac{\text{arctanh}(r_{\text{within}}) + \text{arctanh}(r_{\text{between}})}{2}\right) \tag{4.5}$$

The composite correlation $r_{\text{composite}}$ serves as the primary performance score throughout this work, as it also serves as the main criterion in the official task. To account for variance introduced by random weight initialization and training sequence, each experiment is run three times with fixed seeds (42,123, 456), and all reported scores represent the mean $\pm$ standard deviation across these three runs.

4.2. Temporal Features

The following section explains how temporal context is incorparated to improve the results.

Building on the baseline architecture, temporal context is added following the text-prepending strategy of Ninalga (2023), in which temporal information is encoded via natural language, by prepending it directly to the input text. This approach requires no architextual modifications, as the modified input text is processed by the same tokenizer and attention mechanism as the raw textl.

4.2.1. Prepending Strategies

Five temporal configurations are evaluated, differing in which temporal information is incorporated:

- **None:** No temporal prefix is added; the raw text is used as-is, corresponding to the baseline configuration.
- **Wave:** The collection wave is prepended in the format `"wave: {collection_phase}"`.
- **Date:** The full date is prepended in the format `"year: {Y} month: {m} day: {d}"`.
- **Hour:** The hour of day is prepended in the format `"hour: {h}"`.
- **Full:** Date and hour are prepended jointly, combining the *Date* and *Hour* formats.

These configurations correspond with the temporal signals identified in the Section 3.3: the long-term collection-wave and the short-term. All other components of the model remain identical to the regression formulation of the baseline, isolating the effect on the model performance of the temporal context.

4.2.2. Results

Table 4.2 reports the composite correlation for each temporal configuration, averaged over three seeds (42, 123, 456). All temporal configurations outperform the no-prefix baseline, with the *date* configuration achieving the strongest composite performance ($r_{\text{composite}} = 0.600 \pm 0.005$), followed closely by *wave* (0.599 ± 0.004). Notably, *date* achieves the highest arousal score across all configurations ($r_{\text{arousal}} = 0.538 \pm 0.011$), while *wave* achieves the highest valence score ($r_{\text{valence}} = 0.668 \pm 0.007$). Combining both signals in the *full* configuration does not yield further improvement over *date* or *wave* alone ($r_{\text{composite}} = 0.591 \pm 0.008$), suggesting redundancy between the two temporal signals rather than complementary information. Based on these results, the *date* configuration is selected for all subsequent experiments involving temporal context.

Temporal Mode	$r_{\text{composite}}$	r_{valence}	r_{arousal}
None (baseline)	0.583 ± 0.016	0.645 ± 0.004	0.522 ± 0.034
Wave	0.599 ± 0.004	$\mathbf{0.668 \pm 0.007}$	0.529 ± 0.011
Date	$\mathbf{0.600 \pm 0.005}$	0.662 ± 0.009	$\mathbf{0.538 \pm 0.011}$
Hour	0.587 ± 0.008	0.652 ± 0.016	0.521 ± 0.012
Full (Date + Hour)	0.591 ± 0.008	0.656 ± 0.014	0.527 ± 0.025

Table 4.2.: Composite correlation for each temporal prepending strategy, averaged over seeds 42, 123, 456. Best values per column in bold.

4.3. User-specific Features

The following section explains how user-specific features are incorparated to improve the results.

Independently from the temporal context introduced in Section 4.2, user identity is incorporated following the UserIdentifier approach following Mireshghallah et al. (2022). Rather than prepending human-readable text, a unique sequence of L randomly sampled token as IDs, drawn uniformly from the tokenizer's full vocabulary, is generated for each user and inserted directly after the `[CLS]` token, before the tokenized input text. Crucially, the user identifier tokens share the same embedding parameters as the input text no additional user-specific parameters or model extensions are introduced. Mireshghallah et al. (2022) show that this parameter-tied, randomly sampled variant outperforms both structured identifier types (e.g., digits or non-alphanumeric tokens) and a prefix-tuning baseline by up to 13% in classification accuracy, while requiring no additional training phases per user.

4.3.1. Rare User Handeling

Since a substantial number of users contributes only a small number of texts, (see Section 3.3.4)), , users with fewer than `MIN_USER_TEXTS` entries in the training data are mapped to a shared placeholder identity, `UNKNOWN_USER`, rather than receiving an individual token sequence. This prevents the model from learning unreliable, low-sample user representations and ensures that every user identifier occurring at inference time, including previously unseen users, maps to a well-defined token sequence.

4.3.2. Configuration

Two hyperparameters determine the user-aware configuration: the minimum number of texts required for a user to receive an individual identifier (`MIN_USER_TEXTS`), and the length of the randomly sampled token sequence per user (`USER_ID_LENGTH`, denoted L in Mireshghallah et al. (2022)). Both hyperparameters are optimized sequentially over three seeds (42, 123, 456), applied on top of the regression baseline described in Section 4.1, without temporal features: `MIN_USER_TEXTS` is first optimized with `USER_ID_LENGTH` fixed at its default value, after which `USER_ID_LENGTH` is optimized with `MIN_USER_TEXTS` fixed at its selected value of 15. The combination of user-aware and temporal features is evaluated separately in Section 4.4.

4.3.3. Results

Table 4.3 reports the composite correlation for the evaluated configurations. The final configuration, `MIN_USER_TEXTS` = 15 and `USER_ID_LENGTH` = 2, achieves the strongest composite performance ($r_{\text{composite}} = 0.596 \pm 0.008$). Compared to the minimum-texts threshold alone, reducing the identifier length to $L = 2$ tokens yields a slightly higher composite score and valence correlation ($r_{\text{valence}} = 0.660 \pm 0.020$) at a marginally lower arousal score ($r_{\text{arousal}} = 0.532 \pm 0.003$), indicating a trade-off between the two target dimensions as identifier length varies. This is broadly consistent with Mireshghallah et al. (2022), who report that shorter identifier sequences preserve more of the available context window for the textual content itself.

Configuration	$r_{\text{composite}}$	r_{valence}	r_{arousal}
MIN_USER_TEXTS = 15	0.595 ± 0.004	0.654 ± 0.006	$\mathbf{0.536 \pm 0.002}$
USER_ID_LENGTH = 2	$\mathbf{0.596 \pm 0.008}$	$\mathbf{0.660 \pm 0.020}$	0.532 ± 0.003

Table 4.3.: Composite correlation for the user-aware ablation, averaged over seeds 42, 123, 456. Best values per column in bold.

4.4. Combined Temporal and User-Aware Features

After evaluating temporal and user-aware features independently in Sections 4.2 and 4.3, the following section examines their combination. Both features are applied together. Both best configuration were chosen for the combined evaluation. As both strategies add further complexity to the input sequence, the full hyperparameter search is reapeated, including pooling strategy and head hidden dimension.

4.4.1. Results

Table 4.4 reports the composite correlation for the combined configuration. The best performing setup: mean pooling, a head hidden dimension of 256, learning rate 2×10^{-5}, batch size 16, and dropout 0.1, achieves $r_{\text{composite}} = 0.609 \pm 0.010$, exceeding both the temporal-only configuration (0.600 ± 0.005, Section 4.2) and the user-aware-only configuration (0.596 ± 0.008, Section 4.3). This improvement is most pronounced for arousal ($r_{\text{arousal}} = 0.544 \pm 0.016$), which exceeds the best arousal score achieved by either feature individually, while valence reaches its highest score across all configurations evaluated so far ($r_{\text{valence}} = 0.681 \pm 0.002$ under the initial encoder/pooling configuration, though the final selected configuration achieves 0.674 ± 0.018). These results indicate that temporal and user-aware context provide complementary, rather than redundant, predictive signal when combined, in contrast to the redundancy observed between the two temporal sub-signals in Section 4.2.

Parameter (winner)	$r_{\text{composite}}$	r_{valence}	r_{arousal}
Encoder = deberta-base-mnli	0.603 ± 0.004	**0.681 ± 0.002**	0.525 ± 0.010
Pooling = mean	0.603 ± 0.004	0.681 ± 0.002	0.525 ± 0.010
Head Hidden = 256	**0.609 ± 0.010**	0.674 ± 0.018	**0.544 ± 0.016**
Learning Rate = 2e-05	0.609 ± 0.010	0.674 ± 0.018	0.544 ± 0.016
Batch Size = 16	0.609 ± 0.010	0.674 ± 0.018	0.544 ± 0.016
Dropout = 0.1	0.609 ± 0.010	0.674 ± 0.018	0.544 ± 0.016

Table 4.4.: Sequential hyperparameter ablation for the combined temporal and user-aware configuration, averaged over seeds 42, 123, 456. Each row reports the composite correlation after the respective parameter's winning value is fixed; unchanged values across rows indicate no further improvement at that step.

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A. Additional Experimental Results

A.1. Ablation Studies for the Baseline

B. Proof Details

Ehrenwörtliche Erklärung

Ich versichere, dass ich die beiliegende Bachelor-, Master-, Seminar-, oder Projektarbeit ohne Hilfe Dritter und ohne Benutzung anderer als der angegebenen Quellen und in der untenstehenden Tabelle angegebenen Hilfsmittel angefertigt und die den benutzten Quellen wörtlich oder inhaltlich entnommenen Stellen als solche kenntlich gemacht habe. Diese Arbeit hat in gleicher oder ähnlicher Form noch keiner Prüfungsbehörde vorgelegen. Ich bin mir bewusst, dass eine falsche Erklärung rechtliche Folgen haben wird.

Declaration of Used AI Tools

Tool	Purpose	Where?	Useful?
ChatGPT	Rephrasing	Throughout	+
DeepL	Translation	Throughout	+
ResearchGPT	Summarization of related work	Sec. **??**	-
Dall-E	Image generation	Figs. 2, 3	++
GPT-4	Code generation	functions.py	+
ChatGPT	Related work hallucination	Most of bibliography	++

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